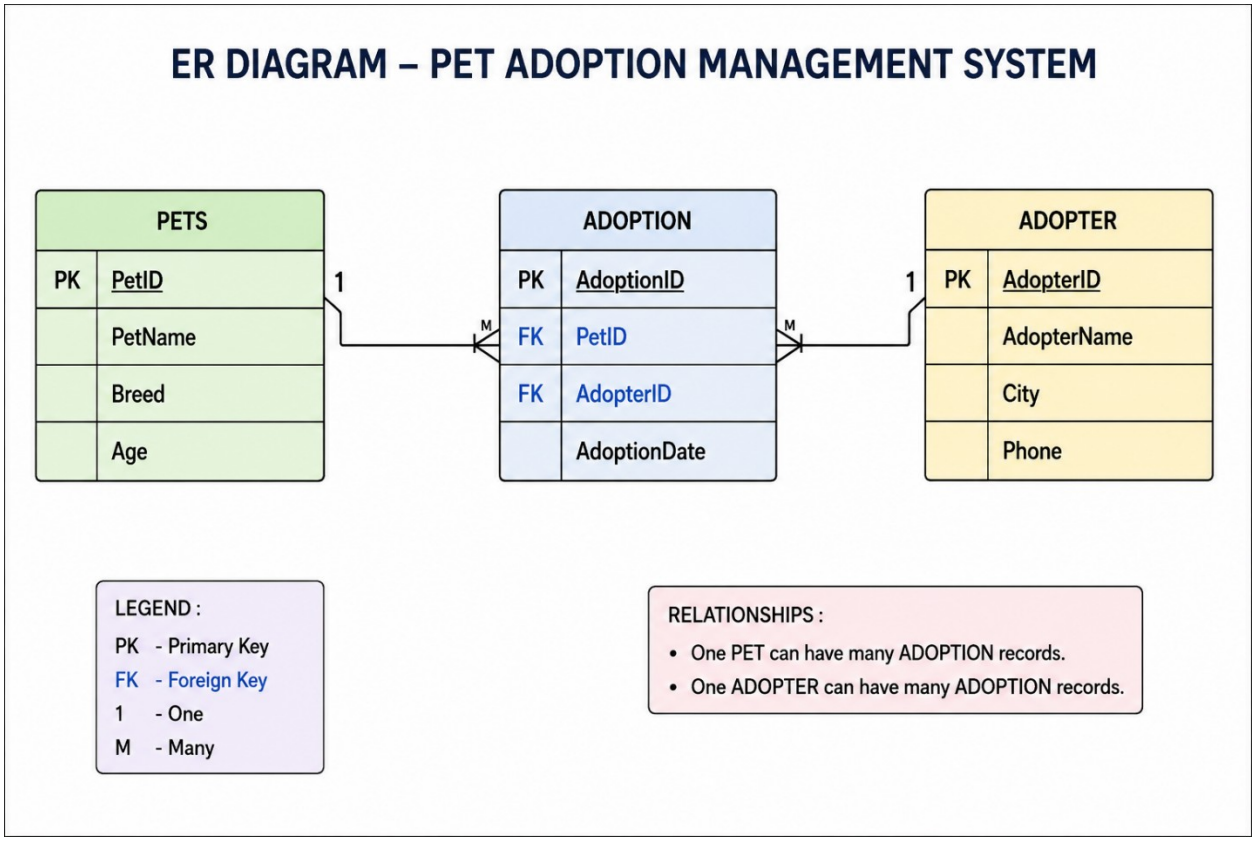


The ER diagram shows the database structure of a Pet Adoption Management System. It includes three entities: Pets, Adoption, and Adopter. The Pets table stores pet details, while the Adopter table stores adopter information. The Adoption table connects pets with adopters and records the adoption date. Primary and foreign keys are used to link the tables. This design helps organize data and makes the adoption process easy to manage.

Database Name : Finalpro

Creating table: Pets

```
mysql> CREATE DATABASE Finalpro;
Query OK, 1 row affected (0.01 sec)

mysql> USE finalpro;
Database changed
mysql> CREATE TABLE Pets
    -> (
    ->     PetID INT PRIMARY KEY,
    ->     PetName VARCHAR(30) NOT NULL,
    ->     PetType VARCHAR(30) NOT NULL,
    ->     Age INT CHECK(Age>0)
    -> );
Query OK, 0 rows affected (0.02 sec)
```

Desc: Pets

inserted values in: Pets

```
mysql> DESC Pets;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| PetID   | int         | NO   | PRI | NULL    |       |
| PetName | varchar(30) | NO   |     | NULL    |       |
| PetType | varchar(30) | NO   |     | NULL    |       |
| Age     | int         | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
4 rows in set (0.01 sec)

mysql> INSERT INTO Pets
    -> VALUES
    -> (101,'Buddy','Golden Retriever',2),
    -> (102,'Luna','Persian Cat',1),
    -> (103,'Rocky','Labrador Puppy',3),
    -> (104,'Bella','Indian Dog',2),
    -> (105,'Mithu','Parrot',1),
    -> (106,'Snow','Rabbit',1),
    -> (107,'Max','German Shepherd',4),
    -> (108,'Daisy','Pomeranian',2),
    -> (109,'Coco','Persian Cat',3),
    -> (110,'Tiger','Indian Dog',5);
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Successfully inserted values in: Pets

Select * query for : Pets

```
mysql> SELECT * FROM Pets;
+-------+---------+------------------+------+
| PetID | PetName | PetType          | Age  |
+-------+---------+------------------+------+
|   101 | Buddy   | Golden Retriever |    2 |
|   102 | Luna    | Persian Cat      |    1 |
|   103 | Rocky   | Labrador Puppy   |    3 |
|   104 | Bella   | Indian Dog       |    2 |
|   105 | Mithu   | Parrot           |    1 |
|   106 | Snow    | Rabbit           |    1 |
|   107 | Max     | German Shepherd  |    4 |
|   108 | Daisy   | Pomeranian       |    2 |
|   109 | Coco    | Persian Cat      |    3 |
|   110 | Tiger   | Indian Dog       |    5 |
+-------+---------+------------------+------+
10 rows in set (0.00 sec)
```

Showing all Pets

Creating table: Adopters

```
mysql> CREATE TABLE Adopters
    -> (
    ->     AdopterID INT PRIMARY KEY,
    ->     AdopterName VARCHAR(30) NOT NULL,
    ->     Email VARCHAR(50) UNIQUE,
    ->     City VARCHAR(30) NOT NULL
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> DESC Adopters;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| AdopterID   | int         | NO   | PRI | NULL    |       |
| AdopterName | varchar(30) | NO   |     | NULL    |       |
| Email       | varchar(50) | YES  | UNI | NULL    |       |
| City        | varchar(30) | NO   |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Desc Adopters

Successfully inserted values in: Adopters

```
mysql> INSERT INTO Adopters
    -> VALUES
    -> (201,'Vineet','vineet@gmail.com','Badlapur'),
    -> (202,'Rahul','rahul@gmail.com','Mumbai'),
    -> (203,'Sneha','sneha@gmail.com','Thane'),
    -> (204,'Amit','amit@gmail.com','Pune'),
    -> (205,'Neha','neha@gmail.com','Kalyan'),
    -> (206,'Riya','riya@gmail.com','Dombivli'),
    -> (207,'Karan','karan@gmail.com','Nashik'),
    -> (208,'Priya','priya@gmail.com','Panvel'),
    -> (209,'Arjun','arjun@gmail.com','Ulhasnagar'),
    -> (210,'Pooja','pooja@gmail.com','Ambernath');
Query OK, 10 rows affected (0.00 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Adopters;
+-----------+-------------+------------------+------------+
| AdopterID | AdopterName | Email            | City       |
+-----------+-------------+------------------+------------+
|       201 | Vineet      | vineet@gmail.com | Badlapur   |
|       202 | Rahul       | rahul@gmail.com  | Mumbai     |
|       203 | Sneha       | sneha@gmail.com  | Thane      |
|       204 | Amit        | amit@gmail.com   | Pune       |
|       205 | Neha        | neha@gmail.com   | Kalyan     |
|       206 | Riya        | riya@gmail.com   | Dombivli   |
|       207 | Karan       | karan@gmail.com  | Nashik     |
|       208 | Priya       | priya@gmail.com  | Panvel     |
|       209 | Arjun       | arjun@gmail.com  | Ulhasnagar |
|       210 | Pooja       | pooja@gmail.com  | Ambernath  |
+-----------+-------------+------------------+------------+
10 rows in set (0.00 sec)
```

Select * query for: Adopters

Creating table: Adoption

```
mysql> CREATE TABLE Adoption
    -> (
    ->     AdoptionID INT PRIMARY KEY,
    ->     PetID INT,
    ->     AdopterID INT,
    ->     AdoptionDate DATE NOT NULL,
    ->
    ->     FOREIGN KEY(PetID) REFERENCES Pets(PetID),
    ->     FOREIGN KEY(AdopterID) REFERENCES Adopters(AdopterID)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> DESC Adoption;
+--------------+------+------+-----+---------+-------+
| Field        | Type | Null | Key | Default | Extra |
+--------------+------+------+-----+---------+-------+
| AdoptionID   | int  | NO   | PRI | NULL    |       |
| PetID        | int  | YES  | MUL | NULL    |       |
| AdopterID    | int  | YES  | MUL | NULL    |       |
| AdoptionDate | date | NO   |     | NULL    |       |
+--------------+------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Desc Adoption

Successfully inserted values in: Adoption

```
mysql> INSERT INTO Adoption
    -> VALUES
    -> (301,101,201,'2026-05-18'),
    -> (302,102,202,'2026-05-27'),
    -> (303,103,203,'2026-06-08'),
    -> (304,104,204,'2026-06-19'),
    -> (305,105,205,'2026-06-29'),
    -> (306,106,206,'2026-07-06'),
    -> (307,107,207,'2026-07-08'),
    -> (308,108,208,'2026-07-12'),
    -> (309,109,209,'2026-07-15'),
    -> (310,110,210,'2026-07-20');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Adoption;
+------------+-------+-----------+--------------+
| AdoptionID | PetID | AdopterID | AdoptionDate |
+------------+-------+-----------+--------------+
|        301 |   101 |       201 | 2026-05-18   |
|        302 |   102 |       202 | 2026-05-27   |
|        303 |   103 |       203 | 2026-06-08   |
|        304 |   104 |       204 | 2026-06-19   |
|        305 |   105 |       205 | 2026-06-29   |
|        306 |   106 |       206 | 2026-07-06   |
|        307 |   107 |       207 | 2026-07-08   |
|        308 |   108 |       208 | 2026-07-12   |
|        309 |   109 |       209 | 2026-07-15   |
|        310 |   110 |       210 | 2026-07-20   |
+------------+-------+-----------+--------------+
10 rows in set (0.00 sec)
```

Select * query for: Adoption

Fetching Pets by Age

```
mysql> SELECT * FROM Pets
    -> WHERE Age=2;
+-------+---------+------------------+------+
| PetID | PetName | PetType          | Age  |
+-------+---------+------------------+------+
|   101 | Buddy   | Golden Retriever |    2 |
|   104 | Bella   | Indian Dog       |    2 |
|   108 | Daisy   | Pomeranian       |    2 |
+-------+---------+------------------+------+
3 rows in set (0.00 sec)
```

Fetching pets by Age and PetType

```
mysql> SELECT * FROM Pets
    -> WHERE Age>=2 AND PetType='Indian Dog';
+-------+---------+------------+------+
| PetID | PetName | PetType    | Age  |
+-------+---------+------------+------+
|   104 | Bella   | Indian Dog |    2 |
|   110 | Tiger   | Indian Dog |    5 |
+-------+---------+------------+------+
2 rows in set (0.00 sec)
```

Fetching Adopters by City

```
mysql> SELECT * FROM Adopters
    -> WHERE City='Mumbai' OR City='Pune';
+-----------+-------------+-----------------+--------+
| AdopterID | AdopterName | Email           | City   |
+-----------+-------------+-----------------+--------+
|       202 | Rahul       | rahul@gmail.com | Mumbai |
|       204 | Amit        | amit@gmail.com  | Pune   |
+-----------+-------------+-----------------+--------+
2 rows in set (0.00 sec)
```

Fetching Adopters where name starts with “ p “

```
mysql> SELECT * FROM Adopters
    -> WHERE AdopterName LIKE 'P%';
+-----------+-------------+-----------------+-----------+
| AdopterID | AdopterName | Email           | City      |
+-----------+-------------+-----------------+-----------+
|       208 | Priya       | priya@gmail.com | Panvel    |
|       210 | Pooja       | pooja@gmail.com | Ambernath |
+-----------+-------------+-----------------+-----------+
2 rows in set (0.00 sec)
```

Fetching pets between specific Age

```
mysql> SELECT * FROM Pets
    -> WHERE Age BETWEEN 2 AND 4;
+-------+---------+------------------+------+
| PetID | PetName | PetType          | Age  |
+-------+---------+------------------+------+
|   101 | Buddy   | Golden Retriever |    2 |
|   103 | Rocky   | Labrador Puppy   |    3 |
|   104 | Bella   | Indian Dog       |    2 |
|   107 | Max     | German Shepherd  |    4 |
|   108 | Daisy   | Pomeranian       |    2 |
|   109 | Coco    | Persian Cat      |    3 |
+-------+---------+------------------+------+
6 rows in set (0.00 sec)
```

Fetching Adopters from particular cities

```
mysql> SELECT * FROM Adopters
    -> WHERE City IN ('Mumbai','Pune','Thane');
+-----------+-------------+-----------------+--------+
| AdopterID | AdopterName | Email           | City   |
+-----------+-------------+-----------------+--------+
|       202 | Rahul       | rahul@gmail.com | Mumbai |
|       203 | Sneha       | sneha@gmail.com | Thane  |
|       204 | Amit        | amit@gmail.com  | Pune   |
+-----------+-------------+-----------------+--------+
3 rows in set (0.00 sec)
```

Sorting Pets by age

```
mysql> SELECT * FROM Pets
    -> ORDER BY Age;
+-------+---------+------------------+------+
| PetID | PetName | PetType          | Age  |
+-------+---------+------------------+------+
|   102 | Luna    | Persian Cat      |    1 |
|   105 | Mithu   | Parrot           |    1 |
|   106 | Snow    | Rabbit           |    1 |
|   101 | Buddy   | Golden Retriever |    2 |
|   104 | Bella   | Indian Dog       |    2 |
|   108 | Daisy   | Pomeranian       |    2 |
|   103 | Rocky   | Labrador Puppy   |    3 |
|   109 | Coco    | Persian Cat      |    3 |
|   107 | Max     | German Shepherd  |    4 |
|   110 | Tiger   | Indian Dog       |    5 |
+-------+---------+------------------+------+
10 rows in set (0.00 sec)

mysql> SELECT * FROM Pets
    -> ORDER BY Age DESC;
+-------+---------+------------------+------+
| PetID | PetName | PetType          | Age  |
+-------+---------+------------------+------+
|   110 | Tiger   | Indian Dog       |    5 |
|   107 | Max     | German Shepherd  |    4 |
|   103 | Rocky   | Labrador Puppy   |    3 |
|   109 | Coco    | Persian Cat      |    3 |
|   101 | Buddy   | Golden Retriever |    2 |
|   104 | Bella   | Indian Dog       |    2 |
|   108 | Daisy   | Pomeranian       |    2 |
|   102 | Luna    | Persian Cat      |    1 |
|   105 | Mithu   | Parrot           |    1 |
|   106 | Snow    | Rabbit           |    1 |
+-------+---------+------------------+------+
10 rows in set (0.00 sec)
```

Sorting pets by age in Desc order

Fetching all unique Pet breeds

```
mysql> SELECT DISTINCT PetType
    -> FROM Pets;
+------------------+
| PetType          |
+------------------+
| Golden Retriever |
| Persian Cat      |
| Labrador Puppy   |
| Indian Dog       |
| Parrot           |
| Rabbit           |
| German Shepherd  |
| Pomeranian       |
+------------------+
8 rows in set (0.00 sec)
```

Counting total no of Pets

```
mysql> SELECT COUNT(*) AS TotalPets
    -> FROM Pets;
+-----------+
| TotalPets |
+-----------+
|        10 |
+-----------+
1 row in set (0.01 sec)
```

Fetching minimum and maximum & average age from PeTS

```
mysql> SELECT MAX(Age)
    -> FROM Pets;
+----------+
| MAX(Age) |
+----------+
|        5 |
+----------+
1 row in set (0.00 sec)

mysql> SELECT MIN(Age)
    -> FROM Pets;
+----------+
| MIN(Age) |
+----------+
|        1 |
+----------+
1 row in set (0.00 sec)

mysql> SELECT AVG(Age)
    -> FROM Pets;
+----------+
| AVG(Age) |
+----------+
|   2.4000 |
+----------+
1 row in set (0.00 sec)
```

Updating age of a Pet

```
mysql>
mysql> UPDATE Pets
    -> SET Age=3
    -> WHERE PetID=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Pets;
+-------+---------+-----------------+------+
| PetID | PetName | PetType         | Age  |
+-------+---------+-----------------+------+
|   101 | Buddy   | Golden Retriever |    2 |
|   102 | Luna    | Persian Cat      |    3 |
|   103 | Rocky   | Labrador Puppy   |    3 |
|   104 | Bella   | Indian Dog       |    2 |
|   105 | Mithu   | Parrot           |    1 |
|   106 | Snow    | Rabbit           |    1 |
|   107 | Max     | German Shepherd  |    4 |
|   108 | Daisy   | Pomeranian       |    2 |
|   109 | Coco    | Persian Cat      |    3 |
|   110 | Tiger   | Indian Dog       |    5 |
+-------+---------+-----------------+------+
10 rows in set (0.00 sec)
```

Adding new values into Pets table

```
mysql> INSERT INTO Pets
    -> VALUES (111,'Bruno','Beagle',2);
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM Pets;
+-------+---------+-----------------+------+
| PetID | PetName | PetType         | Age  |
+-------+---------+-----------------+------+
|   101 | Buddy   | Golden Retriever |    2 |
|   102 | Luna    | Persian Cat      |    3 |
|   103 | Rocky   | Labrador Puppy   |    3 |
|   104 | Bella   | Indian Dog       |    2 |
|   105 | Mithu   | Parrot           |    1 |
|   106 | Snow    | Rabbit           |    1 |
|   107 | Max     | German Shepherd  |    4 |
|   108 | Daisy   | Pomeranian       |    2 |
|   109 | Coco    | Persian Cat      |    3 |
|   110 | Tiger   | Indian Dog       |    5 |
|   111 | Bruno   | Beagle           |    2 |
+-------+---------+-----------------+------+
11 rows in set (0.00 sec)

mysql> DELETE FROM Pets
    -> WHERE PetID = 111;
Query OK, 1 row affected (0.00 sec)
```

Deleted new pet values we added from pets

```
mysql> DELETE FROM Pets
    -> WHERE PetID = 111;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Pets;
+-------+---------+-----------------+------+
| PetID | PetName | PetType         | Age  |
+-------+---------+-----------------+------+
|   101 | Buddy   | Golden Retriever |    2 |
|   102 | Luna    | Persian Cat     |    3 |
|   103 | Rocky   | Labrador Puppy  |    3 |
|   104 | Bella   | Indian Dog      |    2 |
|   105 | Mithu   | Parrot          |    1 |
|   106 | Snow    | Rabbit          |    1 |
|   107 | Max     | German Shepherd |    4 |
|   108 | Daisy   | Pomeranian      |    2 |
|   109 | Coco    | Persian Cat     |    3 |
|   110 | Tiger   | Indian Dog      |    5 |
+-------+---------+-----------------+------+
10 rows in set (0.00 sec)
```

Updating Pets table schema

```
mysql> ALTER TABLE Pets
    -> ADD Gender VARCHAR(10);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Pets;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| PetID   | int         | NO   | PRI | NULL    |       |
| PetName | varchar(30) | NO   |     | NULL    |       |
| PetType | varchar(30) | NO   |     | NULL    |       |
| Age     | int         | YES  |     | NULL    |       |
| Gender  | varchar(10) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql> UPDATE Pets
    -> SET Gender='Male'
    -> WHERE PetID=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

Adding new values in column we created in Pets table

```
mysql> UPDATE Pets
    -> SET Gender='Male'
    -> WHERE PetID=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM PETS;
+-------+---------+-----------------+------+--------+
| PetID | PetName | PetType         | Age  | Gender |
+-------+---------+-----------------+------+--------+
|   101 | Buddy   | Golden Retriever |    2 | Male   |
|   102 | Luna    | Persian Cat     |    3 | NULL   |
|   103 | Rocky   | Labrador Puppy  |    3 | NULL   |
|   104 | Bella   | Indian Dog      |    2 | NULL   |
|   105 | Mithu   | Parrot          |    1 | NULL   |
|   106 | Snow    | Rabbit          |    1 | NULL   |
|   107 | Max     | German Shepherd |    4 | NULL   |
|   108 | Daisy   | Pomeranian      |    2 | NULL   |
|   109 | Coco    | Persian Cat     |    3 | NULL   |
|   110 | Tiger   | Indian Dog      |    5 | NULL   |
+-------+---------+-----------------+------+--------+
10 rows in set (0.00 sec)
```

MySQL Schema alteration: data type modification

```
mysql> ALTER TABLE Pets
    -> MODIFY Gender VARCHAR(20);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Pets;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| PetID   | int         | NO   | PRI | NULL    |       |
| PetName | varchar(30) | NO   |     | NULL    |       |
| PetType | varchar(30) | NO   |     | NULL    |       |
| Age     | int         | YES  |     | NULL    |       |
| Gender  | varchar(20) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

Dropping column gender from Pets

```
mysql> ALTER TABLE Pets
    -> DROP COLUMN Gender;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC Pets;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| PetID   | int         | NO   | PRI | NULL    |       |
| PetName | varchar(30) | NO   |     | NULL    |       |
| PetType | varchar(30) | NO   |     | NULL    |       |
| Age     | int         | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

Joining tables using Inner join

```
mysql> SELECT
    -> Pets.PetName,
    -> Adopters.AdopterName,
    -> Adoption.AdoptionDate
    -> FROM Adoption
    -> INNER JOIN Pets
    -> ON Adoption.PetID=Pets.PetID
    -> INNER JOIN Adopters
    -> ON Adoption.AdopterID=Adopters.AdopterID;
+---------+-------------+--------------+
| PetName | AdopterName | AdoptionDate |
+---------+-------------+--------------+
| Buddy   | Vineet      | 2026-05-18   |
| Luna    | Rahul       | 2026-05-27   |
| Rocky   | Sneha       | 2026-06-08   |
| Bella   | Amit        | 2026-06-19   |
| Mithu   | Neha        | 2026-06-29   |
| Snow    | Riya        | 2026-07-06   |
| Max     | Karan       | 2026-07-08   |
| Daisy   | Priya       | 2026-07-12   |
| Coco    | Arjun       | 2026-07-15   |
| Tiger   | Pooja       | 2026-07-20   |
+---------+-------------+--------------+
10 rows in set (0.01 sec)
```

Joining tables using Left join

```
mysql> SELECT
    -> Pets.PetName,
    -> Adopters.AdopterName
    -> FROM Pets
    -> LEFT JOIN Adoption
    -> ON Pets.PetID=Adoption.PetID
    -> LEFT JOIN Adopters
    -> ON Adoption.AdopterID=Adopters.AdopterID;
+---------+-------------+
| PetName | AdopterName |
+---------+-------------+
| Buddy   | Vineet      |
| Luna    | Rahul       |
| Rocky   | Sneha       |
| Bella   | Amit        |
| Mithu   | Neha        |
| Snow    | Riya        |
| Max     | Karan       |
| Daisy   | Priya       |
| Coco    | Arjun       |
| Tiger   | Pooja       |
+---------+-------------+
10 rows in set (0.00 sec)
```

Joining tables using Right join

```
mysql> SELECT
    -> Pets.PetName,
    -> Adopters.AdopterName
    -> FROM Pets
    -> RIGHT JOIN Adoption
    -> ON Pets.PetID=Adoption.PetID
    -> RIGHT JOIN Adopters
    -> ON Adoption.AdopterID=Adopters.AdopterID;
+---------+-------------+
| PetName | AdopterName |
+---------+-------------+
| Buddy   | Vineet      |
| Luna    | Rahul       |
| Rocky   | Sneha       |
| Bella   | Amit        |
| Mithu   | Neha        |
| Snow    | Riya        |
| Max     | Karan       |
| Daisy   | Priya       |
| Coco    | Arjun       |
| Tiger   | Pooja       |
+---------+-------------+
10 rows in set (0.00 sec)
```